In [19]:
```
# 1.
with open("student.txt","r") as f:
    names = f.read().splitlines()

print(names)
```

```
['Shekhar', 'Sagar', 'Dipak', 'Priyanka', 'Roshni', 'Suraj', 'Akash']
```

In [20]:
```
# 2.
with open("student.txt") as f:
    names=f.read().splitlines()

for name in names:
    if len(name) > 5:
        print(name)
```

```
Shekhar
Priyanka
Roshni
```

In [21]:
```
# 3.
with open("student.txt") as f:
    names=f.read().splitlines()

count=sum(name[0].lower() in "aeiou" for name in names)
print(count)
```

```
1
```

In [22]:
```
# 4.
with open("student.txt") as f:
    names=f.read().splitlines()

with open("filtered.txt","w") as f:
    f.write("\n".join(name for name in names if len(name) > 5))
```

In [23]:
```
# 5.
with open("student.txt") as f:
    names=f.read().splitlines()

for name in sorted(names):
    print(name)
```

```
Akash
Dipak
Priyanka
Roshni
Sagar
Shekhar
Suraj
```

In [24]:
```
# 6.
with open("student.txt") as f:
```

```
    names=f.read().splitlines()

for name in names:
    if "a" in name.lower():
        print(name)
```

```
Shekhar
Sagar
Dipak
Priyanka
Suraj
Akash
```

In [25]:
```
# 7.
with open("student.txt") as f:
    names=f.read().splitlines()

freq={}
for name in names:
    freq[name]=freq.get(name,0)+1

print(freq)
```

```
{'Shekhar': 1, 'Sagar': 1, 'Dipak': 1, 'Priyanka': 1, 'Roshni': 1, 'Su
raj': 1, 'Akash': 1}
```

In [26]:
```
# 8.
with open("student.txt") as f:
    names=f.read().splitlines()

for name in names:
    print(name,len(name),name[0],name[-1],name.lower()==name.upper()[
```

```
Shekhar 7 S r False
Sagar 5 S r False
Dipak 5 D k False
Priyanka 8 P a False
Roshni 6 R i False
Suraj 5 S j False
Akash 5 A h False
```

In [27]:
```
# 9.
with open("student.txt") as f:
    names=f.read().splitlines()

names=sorted(names,key=len)

with open("processed.txt","w") as f:
    f.write("\n".join(names))
```

In [28]:
```
#10. Creates files such as A_names.txt, B_names.txt, and S_names.txt
with open("student.txt") as f:
    names=f.read().splitlines()
```

```
groups={}
for name in names:
    letter=name[0].upper()
    groups.setdefault(letter,[]).append(name)

for letter,group in groups.items():
    with open(f"{letter}.txt","w") as f:
        f.write("\n".join(group))
```

In [ ]: